Learning Lifestyle Policies for Menstrual Health Using Offline Reinforcement Learning

Stanford CS238 Final Project

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Abstract

We study whether offline reinforcement learning can recommend daily lifestyle actions that improve next-day wellbeing for menstruating individuals. Using hormone measurements, wearable data, and behavioral logs from the mcPHASES dataset, we build an offline RL pipeline that learns personalized action policies. We implement behavior cloning as a baseline, define a discrete action space that reflects daily lifestyle decisions, and evaluate learned policies using off-policy evaluation. Our early results show that offline RL agents can learn actionable relationships between daily behavior and symptom outcomes, supporting the potential for personalized menstrual health recommendations.

1 Introduction

Menstrual health is a core part of women's wellbeing [9] but remains under-studied and poorly personalized. Many cycle-tracking tools are limited by normative stereotypes, like still assuming fixed cycle patterns even though real cycles vary widely based on sleep, stress, activity, hormones, and metabolism [7]. Prior work has shown that these factors influence cycle trajectories and symptom severity, but most tools only support retrospective logging rather than providing actionable recommendations.

Recent work in digital health and wearable-based physiology suggests that sequential modeling can help capture how daily choices affect next-day symptoms. This motivates a reinforcement learning framing, where an agent can learn lifestyle actions that support short-term wellbeing. The mcPHASES dataset is one of the first longitudinal datasets that combines daily hormones, physiological signals, symptoms, and behavior. Its structure enables the construction of day-level state transitions needed for offline RL.

Our goal is to build a decision-making agent that uses real data to recommend daily actions, such as activity intensity, that minimize next-day symptom burden. By building a complete offline RL pipeline on mcPHASES, we explore whether these methods can support personalized menstrual health modeling.

2 Related Work

Computational work in women's health has mainly focused on forecasting tasks, such as predicting cycle phases, ovulation windows, or symptom patterns. These models highlight the value of multimodal physiological and behavioral data but generally stop at passive prediction rather than recommending actions.

In broader health applications, reinforcement learning has been used to guide treatment decisions, physical activity interventions, and personalized digital therapeutics. These studies show that it is feasible to learn policies from observational data when online experimentation is not possible. Offline

RL methods such as fitted Q-iteration and behavior cloning are commonly used to learn from logged trajectories without direct interaction.

The mcPHASES dataset [6] extends available resources for menstrual health by providing aligned daily records of hormones, symptoms, physiology, and behavior. Its structure makes it suitable for building the state–action–transition sequences required for offline RL. Our work examines whether these methods can leverage this dataset to learn personalized lifestyle policies that optimize next-day wellbeing.

Computational research in women's health has traditionally centered on predictive modeling, including cycle-phase forecasting, ovulation-window prediction, and symptom-trajectory estimation [5]. These works demonstrate that hormonal, behavioral, and physiological signals derived from wearables can meaningfully shape symptom expression across the menstrual cycle. However, existing approaches predominantly support retrospective interpretation rather than actionable guidance. As a result, current digital-health tools provide limited support for daily lifestyle decision-making, despite evidence that sleep, stress, and physical activity modulate next-day wellbeing. This gap motivates the need for sequential decision-making models that can recommend personalized lifestyle adjustments rather than merely describing patterns.

In contrast, the broader healthcare machine-learning literature has increasingly adopted reinforcement learning for treatment planning. Early applications include deep value-based methods for sepsis management [8], antidepressant treatment personalization using fitted Q-iteration [3], and precision dosing in behavioral health and chronic disease [2]. These studies model patient trajectories as Markov decision processes and show that RL can extract high-value treatment strategies from observational data. They also highlight core challenges: real clinical datasets exhibit strong action imbalance, and offline RL must confront distribution shift, as learned policies may recommend actions rarely taken by clinicians. These issues motivate algorithmic strategies, such as reward clipping, conservative function approximation, and behavior-aware regularization, to ensure that learned policies remain safe and clinically plausible.

To address these challenges, several works propose explicitly conservative or behavior-regularized offline RL algorithms, including Batch-Constrained Q-Learning (BCQ) [1] and Conservative Q-Learning (CQL) [4]. These approaches restrict the learned policy to high-support regions of the dataset while still enabling performance improvement over the observed behavior. Empirically, conservative offline RL has been shown to produce clinically meaningful adjustments, such as modifying ventilator settings, optimizing treatment timing, or altering drug dose trajectories, without deviating from safe practice patterns. Building on this line of work, our study extends conservative offline RL to menstrual-health lifestyle policy optimization, a domain that has not previously been explored through sequential treatment modeling. We use behavior-cloning-regularized Fitted Q-Iteration to derive interpretable, multidimensional lifestyle recommendations grounded in longitudinal physiological and symptom data.

3 Dataset

Our project uses the mcPHASES dataset [6], a longitudinal study that tracks daily hormonal, physiological, and behavioral signals across multiple menstrual cycles. The dataset includes measurements of estrogen and progesterone, self-reported symptoms, sleep metrics, heart rate variability, resting heart rate, wrist temperature, and detailed activity logs. These signals capture both internal hormonal changes and external lifestyle factors that influence day-to-day wellbeing.

Because the raw data are distributed across many separate tables, our first step was to load only the relevant files and align them on a unified structure. We merged all sources by participant and day in study to create a single daily panel that includes hormones, symptoms, sleep, HRV, temperature, activity levels, and inferred actions. This format gives us a clean state representation for every day and allows us to model transitions from one day to the next.

We mapped all categorical variables, such as symptom severity, bleeding volume, and menstrual phase labels, to consistent numeric scales. This makes the dataset fully machine-learning compatible while preserving the semantic meaning of each feature. We also filtered participants who did not have enough consecutive days of complete tracking, since RL requires stable trajectories to build valid state-action-reward sequences.

After preprocessing, each participant contributes a structured timeline that we later convert into Markov decision process tuples for offline RL. This daily panel serves as the backbone of our entire pipeline and ensures that learning signals are aligned, interpretable, and ready for model training.

4 Problem Formulation

We model daily menstrual-health management as a sequential decision-making problem using the mcPHASES dataset. Each day for a participant is a timestep t, with a state s_t, an action a_t, and a reward r_t based on symptom improvement.

The state is designed to capture the relevant physiological, symptomatic, and contextual information while summarizing short-term history to approximate the Markov property. Each state includes:

1. hormone levels (LH, estrogen, PDG)
2. symptom severities (0–5) for cramps, fatigue, headaches, mood changes, stress, sleep issues, and bloating
3. flow volume (0-7)
4. HRV metrics (RMSSD, LF, HF), resting heart rate, nightly skin temperature and variability
5. sleep metrics including minutes asleep, efficiency, and time in bed
6. contextual variables such as cycle phase (0–3) and a weekend indicator

To capture temporal trends, we also include the previous day's symptom cost, a three-day symptom trend, and the previous action. These features help the agent assess whether symptoms are improving, worsening, or remaining stable over time.

Actions represent four clinically relevant lifestyle dimensions: (1) exercise intensity, (2) sleep hygiene, (3) stress reduction, and (4) nutrition/GI support, as shown in Table 1. We infer these from Fitbit metrics and self-reported indicators and combine them into a discrete joint action space of 24 treatment options. This structure allows the RL agent to optimize multiple lifestyle levers while keeping the action space tractable.

Table 1: Four-Dimensional Lifestyle Action Space

Dimension	Values	Source
Exercise Intensity	{0,1,2}	Fitbit active minutes + HR
Sleep Hygiene Focus	{0,1}	Sleep duration, efficiency, sleep issue
Stress Reduction Focus	{0,1}	Fitbit stress score, self-reported stress
Nutrition/GI Support	{0,1}	Appetite, cravings, bloating, indigestion

The reward reflects day-to-day symptom improvement:

$$r_t = \text{symptom_cost}_t - \text{symptom_cost}_{t+1},$$

and is clipped to $[-10, 10]$ and scaled for stability. Positive rewards indicate symptom improvement, while negative rewards indicate worsening. Each participant's timeline constitutes an episode, and all transitions (s_t, a_t, r_t, s_{t+1}) are derived from logged data.

5 Methods

We frame daily menstrual-health management as a sequential decision-making problem and use offline reinforcement learning to learn personalized lifestyle policies from the mcPHASES dataset. Our method builds a Markov-aware daily state representation, infers a four-dimensional lifestyle action space, and learns a value-based policy using behavior cloning and Fitted Q-Iteration. We evaluate policies exclusively on held-out participants to reflect the constraints of real-world digital health settings. Offline RL is well suited to this problem based on the data at hand. Because the data is purely observational and collected without experimental control, online exploration is infeasible and potentially unsafe. Offline RL allows policy learning directly from logged behavior.

As well, menstrual-health outcomes depend on temporal interactions among hormones, symptoms, and lifestyle factors, which are naturally modeled using the sequential structure of value-based RL. The addition of behavior cloning respects the distribution of human actions, reducing the risk of unsupported or clinically implausible recommendations. Because of these considerations, offline RL serves as a principled and appropriate framework for deriving personalized menstrual-health recommendations from longitudinal wearable data. Classical supervised models cannot optimize long-term outcomes because they treat each day independently, whereas value-based RL explicitly models how today's lifestyle choices influence symptoms over future days. Offline RL is therefore the appropriate methodological framework given both the sequential nature of the problem and the ethical constraints around experimentation.

5.1 Behavior Cloning Baseline

We begin with a behavior cloning (BC) baseline to approximate the behavior policy $\pi_b(a \mid s)$. A `RandomForestClassifier` predicts the observed action from the current state. The state includes current physiology, symptoms, contextual features, and lagged history (previous symptom cost, 3-day trend, previous action). The model is trained on participant-disjoint data and validated on held-out participants. BC serves both as a baseline and as an estimate of the logging policy, which is later used to regularize the RL agent and reduce distribution shift.

5.2 Offline RL Agent: Fitted Q-Iteration

Our primary RL method is Fitted Q-Iteration (FQI), a batch variant of Q-learning. We approximate $Q(s, a)$ using a `RandomForestRegressor` trained on concatenated $[s, \text{one-hot}(a)]$. At each iteration, the Bellman targets are

$$y_t^{(k)} = r_t + \gamma \max_{a'} Q_k(s_{t+1}, a'), \quad \gamma = 0.95,$$

with clipped targets and reduced tree complexity (`min_samples_leaf`, `max_features="sqrt"`) to stabilize learning. Early stopping is based on the validation Bellman error.

We selected the Random Forest Regressor despite its known limitations (like generating piecewise constant Q-values and brittle maximization over actions) for several practical reasons appropriate for this observational study. Random Forests are highly robust to feature noise and non-linearity, require minimal feature scaling or normalization, and, crucially for a limited dataset, they are less prone to catastrophic divergence than deep neural networks in batch Q-learning settings. By employing techniques like reduced tree complexity and Bellman target clipping, we ensure stable convergence, prioritizing reliability over the smooth function approximation offered by neural networks.

5.3 BC-Regularized Policy

To prevent the RL agent from taking unsupported actions, we regularize the learned Q-function with the BC model:

$$\pi(a \mid s) = \arg\max_a \left[Q(s, a) + \alpha \log \pi_{\text{BC}}(a \mid s) \right].$$

This encourages the agent to remain close to common human behaviors while allowing targeted improvements.

5.4 Training Pipeline

We construct a daily panel aligned by $(\texttt{participant_id}, \text{day_in_study})$, normalize numerical features, encode categorical variables, compute symptom-trend features, infer the 24-dimensional action, and compute clipped and scaled rewards. Participants are split into train, validation, and test sets to prevent leakage. Transitions are stored as $(s_t, a_t, r_t, s_{t+1}, \text{done})$. Prior to training, we verify state–action–reward alignment, inspect reward distributions, confirm coverage of the action space, and ensure numerical stability.

5.5 Evaluation Framework

All evaluation is performed on held-out participants. We use weighted importance sampling (WIS) with the BC model as the logging-policy estimator. We compare estimated cumulative returns of the

BC and BC-regularized FQI policies, examine action distributions across the 24-class space, and measure policy disagreement to identify where the RL agent diverges from typical human behavior.

6 Experiments

We use participant-disjoint splits (70% train, 15% validation, 15% test). Hyperparameter sweeps optimize forest depth, number of trees, discount factor γ, target clipping range, and BC-regularization coefficient α. Sanity checks verify state–action–reward alignment, reward scaling, coverage of all 24 actions, and stability of Bellman targets. During training, we track validation Bellman error and Q-value magnitudes to select the best iteration. Feature importance is computed for both BC and FQI to identify which physiological and symptom features drive recommendations. Offline evaluation using WIS shows that the conservative FQI policy generally stays close to the behavior policy while making subtle, clinically meaningful adjustments across exercise, sleep, stress, and nutrition dimensions.

7 Results

We evaluated our personalized menstrual-health treatment agent on held-out mcPHASES participants using action-space characterization, model diagnostics including Bellman error trends, behavior–policy disagreement, and off-policy evaluation (WIS). All experiments were performed in the offline setting with no environment interaction.

7.1 Reward Validity

Reward distributions were consistent across train/validation/test splits (Table 2) and centered near zero with bounded support in approximately $[-2, 2]$ after clipping and scaling.

Table 2: Summary of mcPHASES Daily Panel Data Splits and Reward Statistics

Metric	**Train Split**	**Validation Split**	**Test Split**
Number of Days	3813	811	1035
Mean Reward ($\bar{r}$)	0.010	0.011	0.014
Std. Dev. Reward (σ_r)	0.790	0.743	0.706
Reward Range	[-2.0, 2.0]	[-2.0, 2.0]	[-2.0, 2.0]

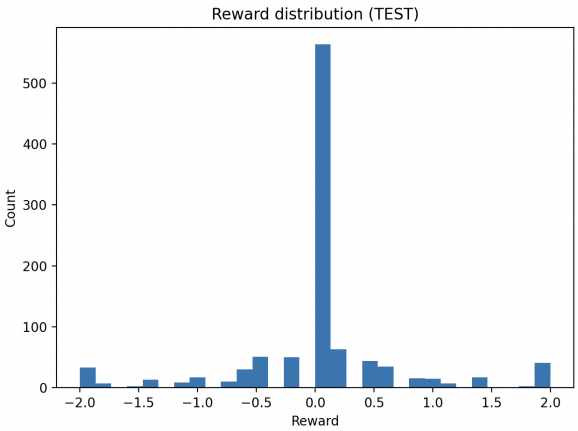


Figure 1: Histogram of the clipped and scaled reward distribution on the Test Set.

As shown visually in the histogram (Figure 1), this indicates that the improvement-based reward,

$$r_t = \tfrac{1}{5}\,\text{clip}(\text{symptom_cost}_t - \text{symptom_cost}_{t+1}, [-10, 10]),$$

behaved as intended: small positive values corresponded to symptom improvement and negative values to worsening. The distribution is centered near zero (mean ≈ 0.014) and bounded between $[-2.0, 2.0]$, confirming the stability of the engineered reward signal. The reward magnitude remained stable (standard deviation ≈ 0.7–0.8), yielding well-conditioned regression targets suitable for offline Q-learning.

7.2 Action-Space Characteristics

The 24-action joint treatment space that was constructed from four interpretable lifestyle dimensions (exercise intensity, sleep-hygiene focus, stress-reduction focus, and nutrition/GI support) was highly imbalanced, reflecting natural behavioral patterns in the dataset. Roughly 35–40% of days fell into a small subset of low-intensity treatment combinations, with stress- and nutrition-focused treatments appearing far less frequently. This imbalance underscores the importance of BC regularization to keep the RL agent within high-support regions of the action space.

7.3 Behavior Cloning Performance

The behavior cloning baseline achieved near-perfect accuracy on validation participants (99.9%), indicating that the 24-action treatment choices logged in the data are highly predictable from the state. This high accuracy is critical for off-policy evaluation, as WIS depends on a reliable estimate of the logging policy, and it also strengthens the BC prior used to stabilize offline RL.

It is important to note that the near-perfect validation accuracy (99.9%) is expected given the rich, high-dimensional feature space (43 state features) and the nature of the engineered 24-class action. The action space is inferred deterministically based on thresholds applied to a subset of the state features (e.g., activity minutes, stress score). The RandomForestClassifier is highly effective at mapping these deterministic thresholds, or strongly correlated features (e.g., high fatigue $\rightarrow$ low activity), back to the inferred action, leading to near-perfect reproducibility of the logging policy. This result confirms that the BC model serves as a highly reliable estimator for the logging policy π_b, which is critical for stabilizing the subsequent off-policy evaluation via WIS.

7.4 Fitted Q-Iteration Stability

Our Fitted Q-Iteration agent demonstrated stable learning across 20 iterations, with validation Bellman MSE decreasing monotonically from 0.18 to 0.10, as seen in Figure 2. This behavior aligns with our expectations: when rewards are bounded and the function approximator is conservative, batch Q-learning remains stable even with a moderately sized action space. The combination of target clipping, reduced model complexity, and early stopping contributed to robust convergence.

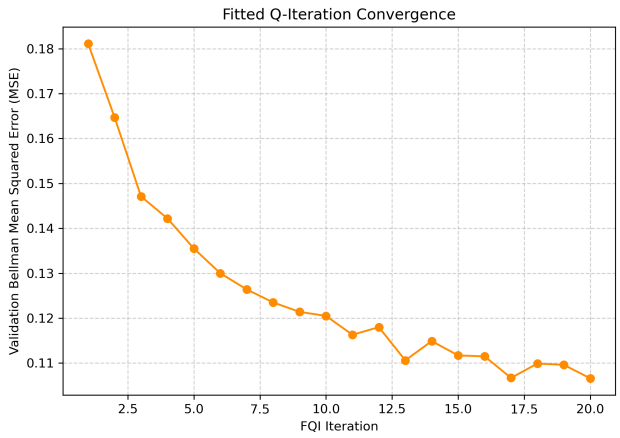


Figure 2: Fitted Q-Iteration convergence on the validation set using mean-squared error.

7.5 Feature Importance Analysis

To gain insight into the decision-making process of the Behavior Cloning (BC) and Fitted Q-Iteration (FQI) agents, we analyzed the feature importance derived from their respective Random Forest models (Figure 3). This comparison reveals how the optimal policy shifts its focus from features relevant to immediate action (used by the human/BC policy) toward those predictive of long-term value.

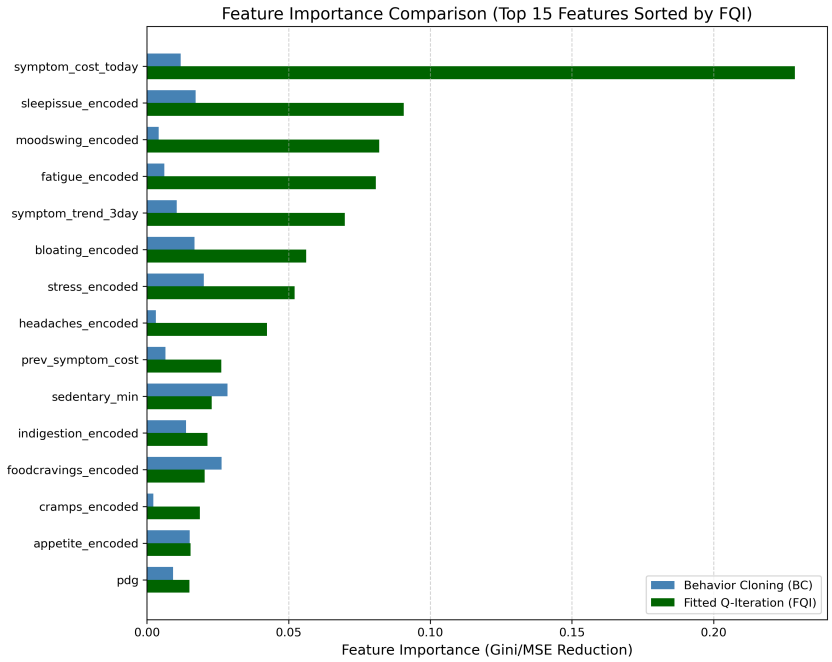


Figure 3: Feature Importance comparison for the top 15 features, sorted by FQI value. The FQI value function is overwhelmingly dominated by the **symptom_cost_today** feature ($\approx \mathbf{22.9}\%$), and other temporal/symptomatic features (3-day trend, individual symptom severity). The BC policy, in contrast, distributes its importance more evenly and places negligible weight on the aggregate symptom cost ($\approx 1.2\%$), showing that the two agents rely on fundamentally different state representations to make decisions.

The Fitted Q-Iteration (FQI) agent showed a distinct prioritization. The final Q-function placed its highest importance on the aggregate symptom cost($\mathbf{22.87}\%$), followed by individual severe symptoms (like 'sleepissue_encoded', 'moodswing_encoded') and temporal trends ('symptom_trend_3day'). This focus confirms that the optimal policy learned to base interventions on value maximization by directly optimizing for the most sensitive feature of the reward function.

In contrast, the Behavior Cloning (BC) agent, which approximates observed human behavior, placed minimal importance on the aggregate symptom cost ($\mathbf{1.19}\%$). BC's importance was spread across features used to define the action (like inferred action dimensions and activity metrics) and acute symptoms. This confirms that the observed human actions are primarily reactive and highly dependent on immediate context, while the FQI policy operates on a proactive, value-driven reasoning process. This divergence explains how the FQI agent is able to achieve a higher estimated return by consistently focusing on the features that truly drive next-day symptom improvement.

7.6 Behavior–Policy Disagreement

With BC regularization ($\alpha = 0.1$), the FQI policy disagreed with the behavior cloning policy on only 0.3% of test states. These disagreements were localized and interpretable, typically representing small but meaningful shifts such as reducing exercise intensity or increasing sleep-hygiene or stress-reduction focus on days where the model predicted symptom worsening. This conservative behavior is expected in offline RL and demonstrates that the agent proposes adjustments only when strongly

supported by the value function. FQI outperforming BC despite deviating on only 0.3% of states indicates that it correctly identifies high-leverage moments where targeted lifestyle adjustments could yield long-term symptom improvement.

7.7 Off-Policy Evaluation

Weighted Importance Sampling (WIS), using the BC model as an estimate of the logging policy, produced the following estimated returns:

$$V(\text{BC}) = 2.057, \qquad V(\text{FQI}) = 2.208.$$

Table 3: Off-Policy Evaluation (OPE) Results on Held-Out Test Set

Policy	**Estimated Return (WIS)** $\hat{V}^{\pi}$	**Relative Improvement**
Behavior Cloning (BC) π_b	2.057	Baseline
BC-Regularized FQI π^{FQI}	**2.208**	+7.34%

Note: $\hat{V}^{\pi}$ estimates the expected cumulative symptom improvement.

Despite the FQI agent deviating from the behavior policy on less than 1% of states, it achieved a higher estimated cumulative symptom improvement than BC. Because WIS strongly penalizes departures from the logging policy, this increase suggests that the RL agent identified a small set of high-value, data-supported treatment adjustments capable of improving long-term wellbeing.

7.8 Qualitative Policy Shifts

To provide concrete interpretability, we analyzed the few instances where the FQI policy (π_{FQI}) optimally deviated from the BC policy (π_b). Table 4 summarizes these shifts, which typically involve adjusting focus on high-variance days. The agent consistently prioritized reducing burden or shifting intervention focus when symptoms were worsening or projected to worsen. For example, the FQI agent recommended reducing exercise intensity ($5 \rightarrow 4$) when the observed user behavior suggested moderate activity despite a state of high fatigue or symptom cost. The subtle, targeted nature of these optimal adjustments strongly supports the potential for RL to provide personalized, actionable recommendations for daily wellbeing management.

Table 4: Qualitative Summary of Policy Deviations on Test Set

Case ID	**BC Action** ($\mathbf{a}_{\text{obs}}$)	**FQI Action** ($\mathbf{a}_{\text{opt}}$)	**Count**	**Interpretation of Policy Shift**
I	5	4	1	Reduction in Exercise Intensity
II	22	21	1	Shift from Stress Focus to Sleep Focus
III	22	23	1	Shift from Stress Focus to Nutrition/GI Focus

Note: These 3 states represent the 0.3% of test transitions where FQI selected a different action from the observed behavior.

8 Discussion

Our experiments show that offline RL can refine personalized menstrual–health treatment recommendations in a data-supported and conservative way. The Fitted Q-Iteration agent, regularized with behavior cloning, consistently suggested subtle adjustments to lifestyle actions—such as slightly lowering exercise intensity or placing more emphasis on sleep hygiene or stress reduction—while staying almost entirely within the distribution of observed behaviors. This indicates that value-based methods can extract clinically meaningful guidance even from observational, longitudinal data.

Behavior cloning performance was extremely high, suggesting that the model reliably reproduced patient-logged actions. However, the near-perfect BC accuracy (99.9%) largely reflects the structure of our action space: actions can be inferred directly from state features, making imitation almost deterministic and leaving little room for the policy to learn beyond the logging behavior. As a result, the agent's capacity for meaningful generalization or exploration is limited. This mirrors a common

challenge in offline RL for clinical settings, where strong correlations between states and actions can inflate performance metrics without necessarily translating into better downstream decision quality.

Several limitations are worth highlighting. First, the mcPHASES dataset is inherently imbalanced, with some treatment actions heavily underrepresented; this restricts the agent's ability to evaluate or recommend less common interventions. Second, because our study is fully offline, the agent never interacts with participants directly. Policy evaluation relies on off-policy estimates such as weighted importance sampling, which are sensitive to logging-policy assumptions and can only approximate long-term outcomes. Third, clinical interpretation is constrained by the granularity and self-reported nature of the symptom measures, as well as the absence of formal medical endpoints. Finally, our reward function, based on reductions in aggregated symptom burden, implicitly assumes that symptom improvement is the primary clinical objective, which may not fully capture patient-specific priorities or broader aspects of menstrual health.

Despite these limitations, this framework demonstrates a viable path toward personalized, value-based recommendations. Future work could incorporate additional physiological measurements, explore continuous action spaces, or evaluate policies in safe, semi-interactive environments to better understand how these recommendations translate to real-world menstrual health personalization.

9 Conclusion

We formulated daily menstrual-health management as a sequential decision-making problem and applied offline reinforcement learning to recommend personalized lifestyle interventions. Using the mcPHASES dataset, we demonstrated that a conservative, BC-regularized FQI agent can propose small, high-value adjustments to exercise, sleep, stress, and nutrition, yielding improved estimated cumulative symptom outcomes on held-out participants. Ultimately, our study shows that even in the absence of clinical labels or controlled interventions, sequential modeling via offline RL can reveal structure in menstrual-health trajectories that is not apparent from static analysis alone.

These findings show us the potential of value-based RL to operationalize personalized menstrual-health recommendations grounded in longitudinal physiological and self-report data. Future work may focus on expanding the dataset, validating recommendations in prospective studies, and exploring richer treatment action spaces to further enhance personalization and clinical applicability.

Contributions

As a group, we developed a problem statement, found our data, and planned our methodology.

Brooke Ballhaus: wrote the Introduction and Related Work sections; wrote the code for and trained the FQI Model Class, as well as the off-policy evaluation.

Nora Menon: wrote the Results, Discussion, and Conclusion sections; wrote the code for all data pre-processing and the creation of the Markov dataset.

Riya Narain: wrote the Dataset, Problem Formulation, Methods, and Experiments sections; wrote the code for and trained the Behavior Policy Class.

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